From symmetry considerations, James Clerk Maxwell showed that the change in electric field also produces a magnetic field which is given by

$$\underbrace{\oint_l \vec{B}\cdot\vec{dl}}_{\text{Magnetic field induced along a closed loop}} = \underbrace{-\frac{d}{dt}\Phi_E}_{\text{Variation of electric flux with time}} = \underbrace{-\frac{d}{dt}\oint_S \vec{E}\cdot\vec{dA}}_{\text{Changing electric flux } \Phi_E \text{ in the region enclosed by the loop}} \quad (5.2)$$

where Φ_E is the electric flux. This is known as **Maxwell's law of induction** which explains that the magnetic field $\vec{B}$ is induced along a closed loop by the changing electric flux Φ_E in the region encircled by that loop. This symmetry between electric and magnetic fields explains the existence of electromagnetic waves such as radio waves, gamma rays, infrared rays etc.

Displacement current – Maxwell's correction

In order to understand how the changing electric field induces magnetic field, let us consider a situation of charging a parallel plate capacitor which contains non–conducting medium between the plates.

Let a time–dependent current i_c, called conduction current be passed through the wire to charge the capacitor.

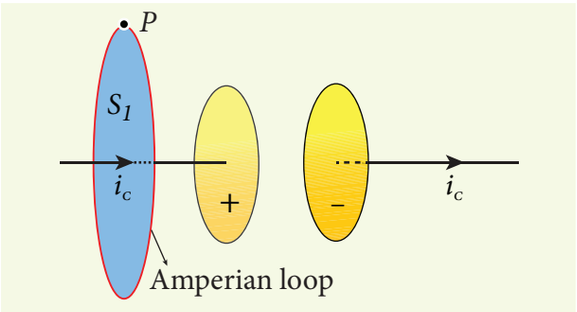


Figure 5.3 Applying Ampere's circuital law - loop enclosing surface

Ampere's circuital law can be used to find the magnetic field produced around the current carrying wire.

To calculate the magnetic field at a point P near the wire and outside the capacitor, let us draw a circular Amperian loop which encloses the circular surface S_1 (Figure 5.3). Using Ampere's circuital law for this loop, we get

$$\oint_{\text{enclosing } S_1} \vec{B}\cdot\vec{dl} = \mu_0 i_C \quad (5.3)$$

where μ_o is the permeability of free space.

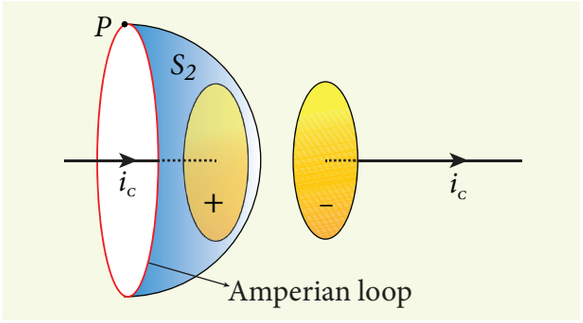


Figure 5.4 Applying Ampere's circuital law - loop enclosing surface S_2

Now, the same loop is enclosed by balloon shaped surface S_2 such that boundaries of two surfaces S_1 and S_2 are same but the shape of the surfaces is different (Figure 5.4). As Ampere's law applied for a given closed loop does not depend on the shape of the enclosing surface, the integrals should give the same answer. But by applying Ampere's circuital law for the surface S_2, we get

$$\oint_{\text{enclosing } S_2} \vec{B}\cdot\vec{dl} = 0 \quad (5.4)$$

The right hand side of equation is zero because the surface S_2 nowhere touches the wire carrying conduction current and further, there is no current flowing between

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